BIOMECHANICAL ASYMMETRY DURING FOOTBALL PASSING FOLLOWING ADDUCTOR INJURY

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The purpose of this study was to investigate whether biomechanical differences exist between the dominant and non-dominant limbs in football players within one month following an adductor injury. Despite the high prevalence of groin injuries in football, limited research has examined lower limb biomechanics during passing specifically, and it remains unknown whether limb asymmetries persist during the early return-to-sport phase. Seven academy football players with a recent adductor injury completed repeated passing drills, while three-dimensional motion capture and musculoskeletal modeling were used to calculate hip and knee joint angles and moments. Significant kinematic asymmetries were observed between limbs, including greater hip adduction on the dominant side ($p < 0.001$), greater hip rotation on the non-dominant side ($p = 0.0013$), and an elevated knee flexion moment on the non-dominant side ($p = 0.006$). These findings highlight the presence of functional asymmetry during football-specific passing after adductor injury and may help inform rehabilitation strategies and return-to-play assessments.

KEYWORDS: football passing, adductor injury, biomechanical asymmetry, lower-limb mechanics, rehabilitation

INTRODUCTION: Groin injuries, particularly those affecting the hip adductor muscles, represent one of the most common diagnoses in football players. Football-specific actions such as passing, sprinting, and changes of direction place substantial loads on the adductor musculature, thereby increasing injury risk (Dupré et al., 2018; Dupré & Potthast, 2020). Despite the high prevalence of these injuries, limited research has examined lower limb biomechanics during the passing task specifically. Previous investigations have predominantly focused on high-intensity activities such as sprinting and change of direction (Dupré, 2019; Dupré & Potthast, 2024), leaving a gap in the understanding of the mechanical demands of passing—a highly frequent football action. Furthermore, it remains unknown whether limb asymmetries persist during the early return-to-sport phase following an adductor injury (Collings et al., 2021). To address this gap, the present study aimed to investigate biomechanical differences between the dominant(DOM) and non-dominant(Non-DOM) limbs in football players within one month after an adductor injury during a football-specific passing task. We hypothesized that significant biomechanical asymmetries would exist between limbs during this phase.

METHODS: A cross-sectional study design was used to investigate lower-limb symmetry in football players one month after an adductor injury. Testing was performed on the training ground of TSG 1899 Hoffenheim, one of Germany's top-ranking football clubs. Seven players with a recent adductor injury were recruited from three different academy age groups. The following parameters were collected: age, body mass, height, injury history, dominant foot, and lower-extremity anthropometrics. Each participant's anthropometric data are presented in Table 1. Each participant completed 30 passes inside a Footballnaut (Beavan et al., 2019). A three-dimensional motion capture system (Vicon™, Oxford, UK) with 12 cameras recording at 240 Hz was used to collect kinematic and kinetic data. Players were allowed to use both DOM and Non-DOM legs freely. The passing motion was defined as the swing phase of the calcaneus from toe-off to ball contact.

Figure 1. Participant characteristics, including height, body mass, age, dominant side, injured side, thigh and shank lengths, and leg length discrepancy.

Player	unit	P02	P32	P34	P53	P64	P74	P88
Height	(m)	1.45	1.596	1.625	1.76	1.758	1.661	1.755
Mass	(kg)	35.9	42.5	50.8	63.2	68.2	62.4	68.9
Age		11	13	13	16	15	14	17
Dominant Side		R	R	R	R	R	L	L
Injury Side		R	R	L	L	R	L	L
Thigh								
Thigh Length Left	(cm)	33.88	38.74	38.61	41.31	43.36	37.58	41.97
Thigh Length Right	(cm)	34.48	38.48	40.53	41.4	42.53	38.83	41.26
Thigh Diff (Left - Right)	(cm)	-0.6	0.26	-1.92	-0.09	0.83	-1.25	0.71
Shank								
Shank Length Left	(cm)	33.34	36.33	35.61	40.5	39.67	35.74	40.87
Shank Length Right	(cm)	30.35	36.72	34.32	40.77	39.2	35.91	39.23
Shank Diff (Left - Right)	(cm)	2.99	-0.39	1.29	-0.27	0.47	-0.17	1.64
Leg Length Diff	(cm)	2.39	-0.13	-0.63	-0.36	1.3	-1.42	2.35

Raw motion capture data were processed using Qualisys software for marker labelling. A musculoskeletal OpenSim model Rajagopal was applied, and inverse kinematics and inverse dynamics were subsequently calculated in MATLAB. The following variables were extracted: hip joint angles and moments (flexion, adduction, rotation) and knee joint angle and moment (flexion).

Passes were excluded from analysis due to marker loss, uneven ground conditions, or ball elevation that interfered with signal processing thresholds. In the SAME group (injured side = dominant side; 5 participants, 127 passes), 19 passes were excluded. In the DIFF group (injured side $\neq$ dominant side; 2 participants, 54 passes), the entire group was excluded due to the limited sample size. After filtering, 108 valid passes remained (87 DOM passes and 21 Non-DOM passes).

The following hypothesis was tested when the dominant leg was the injured leg:

H_0: No systematic difference exists between DOM and Non-DOM limbs (Diff = 0).

H_1: A systematic difference exists (Diff $\neq$ 0).

An alpha level of $\alpha = 0.05$ was used. A significant result led to rejection of H_0 (indicating asymmetry), while a non-significant result indicated symmetry.

All participants, and their legal guardians where applicable, provided written informed consent prior to participation. The study was conducted in accordance with the Declaration of Helsinki and approved by the ethics committee of the German Sport University (No. 125/2016).

RESULTS AND DISCUSSION: Significant kinematic and kinetic asymmetries were observed between the dominant (DOM) and non-dominant (Non-DOM) limbs during the passing motion, with a consistent pattern across three key variables: hip adduction (DOM dominant), hip rotation (Non-DOM dominant), and knee flexion moment (Non-DOM dominant). These findings highlight the presence of inter-limb asymmetry during a sport-specific task and suggest both biomechanical and structural contributors.

The Wilcoxon signed-rank test revealed significant differences in several kinematic variables. (see Table 1)

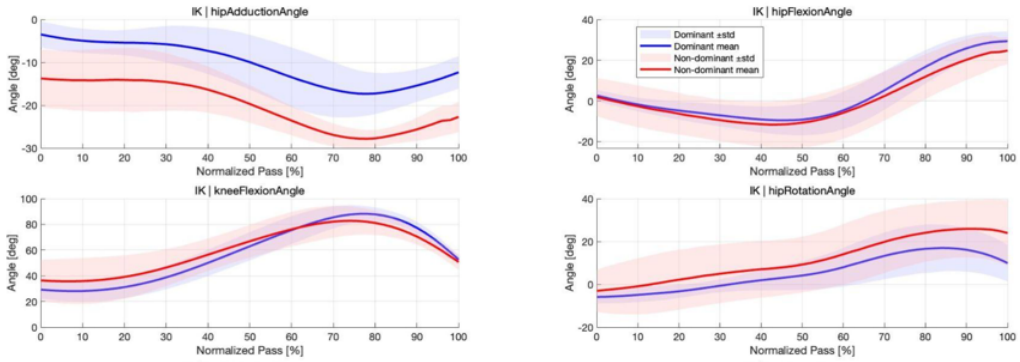
Table 1. Results of the Wilcoxon signed-rank test comparing dominant (DOM) and non-dominant (Non-DOM) limbs across kinematic and kinetic variables.

Variable	Median_diff	Mean_diff	SD_diff	p_value	Significant
hipFlexionAngle	-1.94	2.57	9.76	0.004	**Significant**
hipAdductionAngle	5.78	9.46	8.89	<0.01	**Significant**
hipRotationAngle	0.81	-6.97	14.18	0.0013	**Significant**
kneeFlexionAngle	-5.11	-4.06	11.19	0.097	Not Significant

hipFlexionMoment	0.57	-1.62	9.59	0.53	Not Significant
hipAdductionMoment	-0.77	-13.28	45.10	0.26	Not Significant
hipRotationMoment	3.29	1.81	2.88	0.20	Not Significant
kneeFlexionMoment	-0.37	-3.67	5.27	0.006	**Significant**

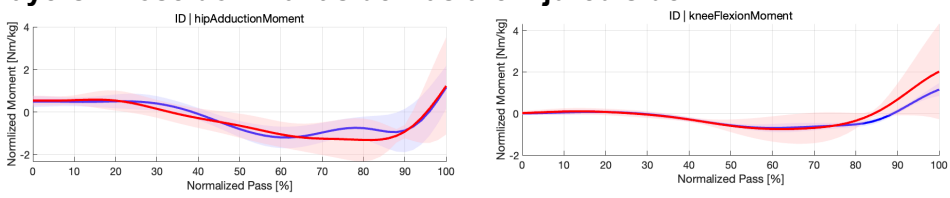
The hip flexion angle differed significantly between limbs ($p = 0.0044$), with the DOM limb reaching a mean peak of 55.19° compared to 33.38° on the Non-DOM side. Hip flexion increased on both sides after approximately 70% of the passing motion. The most pronounced asymmetry was observed in hip adduction ($p < 0.001$), with a mean difference of 9.46°. The DOM limb maintained greater adduction throughout the movement (mean peak: 12.85°), whereas the Non-DOM limb exhibited a mean peak of -1.95°. This pattern may increase medial knee loading and potentially elevate the risk of knee pathologies and genu varum over time.
Hip rotation also differed significantly ($p = 0.0013$), with greater rotation observed in the Non-DOM limb across the entire movement. This may represent a compensatory mechanism to counterbalance increased adduction on the DOM side, shifting load toward the lateral side and potentially contributing to a toe-out gait pattern. (Figure 2)

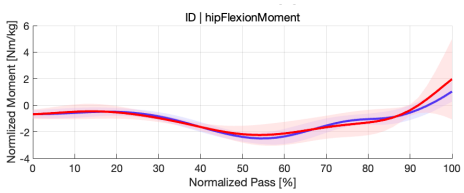
Figure 2. Hip adduction, flexion, and rotation angles during the passing motion for players whose dominant side was the injured side. The dominant (DOM) limb is shown in blue, and the non-dominant (Non-DOM) limb in red. Solid lines represent mean values, and shaded areas indicate standard deviations.

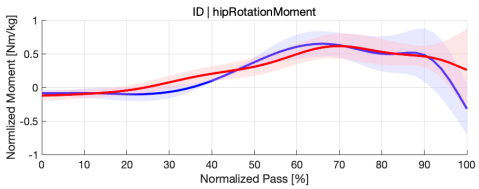


Among kinetic variables, only the knee flexion moment showed a significant difference ($p = 0.006$). After 80% of the passing phase, the Non-DOM limb exhibited a greater and more rapidly increasing knee flexion moment, suggesting a stabilizing or braking function during the terminal phase. No significant differences were observed in hip flexion ($p = 0.5336$), adduction ($p = 0.2632$), or rotation moments ($p = 0.2021$), likely due to high variability and the small sample size. (Figure 3)

Figure 3. Hip adduction, flexion and rotation moment during the passing motion for players whose dominant side was the injured side.







These asymmetries may be partially explained by structural factors such as limb length discrepancy. Previous research (Akodu et al., 2020) indicates that dominant limbs are frequently injured and that many players exhibit some degree of asymmetry. In this study, two of seven players showed discrepancies greater than 20 mm. Although small discrepancies can often be compensated, they may still increase injury risk in athletic populations. Compared to reference values (Lees, 2010), hip and knee flexion angles fell within normal ranges, whereas hip adduction (DOM) and hip rotation (both limbs) exceeded typical values, further supporting the presence of task-specific asymmetry.

From a practical perspective, both external and internal interventions may help address these asymmetries. External strategies include heel lifts, shoe modifications, and custom insoles. Internal strategies should focus on strengthening hip abductors and external rotators, improving pelvic control, balancing muscle strength, enhancing flexibility, and optimizing passing technique.

Several limitations should be acknowledged. The small sample size limits generalizability, and the absence of force plate and electromyographic data restricts interpretation of muscle activity and ground reaction forces. Additionally, angular velocity was not assessed, which may provide further insight in future research.

CONCLUSION: This study demonstrates that following an adductor injury, football players exhibit a characteristic compensatory asymmetry during passing. This includes increased hip adduction in the dominant limb, increased hip rotation in the non-dominant limb, and greater knee flexion moment in the non-dominant limb. These findings suggest a redistribution of joint demands between limbs. From a clinical perspective, rehabilitation should emphasize hip abduction strength, external rotation, and pelvic control to reduce compensatory loading and minimize the risk of subsequent injury.

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